Maximal sieves on the birhizomic stratum site: positive answers for rank-one and valuative monoids

Abstract

We study the adjunction between sheaves on the birhizomic site of a monoid P and sheaves on the trivial site of the face-stratum subcategory B_P. We prove that if every birhizomic covering sieve of every stratum in B_P is maximal, then for every presheaf of abelian groups F on B_P^{op} the adjunction unit

$$F \longrightarrow R\pi_*\pi^{-1}F$$

is a quasi-isomorphism. We verify this maximal-sieve condition for all rank-one monoids and for all finitely generated valuative monoids. Thus the original question has a positive answer in these two regimes. The general non-valuative case and the necessity of the maximal-sieve condition remain open.

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1 Introduction

A *monoid* in this paper is an integral, saturated, sharp, commutative, unital monoid, written additively. For a monoid M, we write M^{gp} for its associated group. The monoid induces a translation-invariant partial order on M^{gp} by

$$g \geq 0 \quad \Longleftrightarrow \quad g \in M.$$

A monoid is *valuative* if this partial order is total.

Given a monoid P, let A_P be the category whose objects are monoid homomorphisms $P \to Q$, with Q again integral, saturated and sharp, and whose morphisms are commutative triangles under P. For $x \in P$, let

$$b_x : P \longrightarrow Q_x := P[-x]^\sharp$$

denote the initial map under P which sends x to 0. Let $B_P \subset A_P$ be the full subcategory on the objects b_x. The opposite category A_P^{op} carries the birhizomic Grothendieck topology, while B_P^{op} carries the trivial topology.

The inclusion $B_P^{\mathrm{op}} \subset A_P^{\mathrm{op}}$ induces, by left Kan extension followed by sheafification, an inverse-image functor

$$\pi^{-1} : \mathrm{Sh}(B_P^{\mathrm{op}}) \longrightarrow \mathrm{Sh}(A_P^{\mathrm{op}}),$$

with right adjoint

$$\pi_* : \mathrm{Sh}(A_P^{\mathrm{op}}) \longrightarrow \mathrm{Sh}(B_P^{\mathrm{op}}).$$

Since the topology on B_P^{op} is trivial, every presheaf on B_P^{op} is a sheaf. The question addressed in this paper is:

> For every presheaf of abelian groups F on B_P^{op}, is the unit
>
> $$F \longrightarrow R\pi_*\pi^{-1}F$$
>
> a quasi-isomorphism?

We prove two positive answers.

- If P has rank one, i.e. $\dim_{\mathbf{Q}}(P^{\mathrm{gp}} \otimes \mathbf{Q}) = 1$, then the answer is yes (Theorem 6.3).
- If P is finitely generated and valuative, then the answer is yes (Theorem 7.2).

The main mechanism is a sufficient condition, denoted (Max), saying that every birhizomic covering sieve of every stratum $b_x \in B_P$ is maximal. Under this condition, sheafification at strata is pointwise trivial, and π_* becomes exact. We then verify (Max) in the two regimes above.

We do not prove the converse. The general non-valuative case remains open; see Section 8.

2 Preliminaries

2.1 Basic conventions

All monoids are assumed integral, saturated, sharp, commutative and unital. For such a monoid M, the group M^{gp} is torsion-free. Indeed, if $ng = 0$ for some $n \geq 1$, then $ng \in M$, so saturation gives $g \in M$; similarly $n(-g) = 0$ gives $-g \in M$; sharpness then forces $g = 0$.

The order on M^{gp} is defined by $g \geq 0$ iff $g \in M$. A monoid is *valuative* if for every $g \in M^{\mathrm{gp}}$, either $g \in M$ or $-g \in M$.

For $x, y \in M^{\mathrm{gp}}$, the original problem writes $x \prec y$ if there are integers m, n such that

$$my \leq x \leq ny.$$

A homomorphism $f : M \to N$ is *infinitesimal* if, whenever $x \in M^{\mathrm{gp}}$, $y \in M$, $f(x) \in N$, $x \prec y$, and $f(y) \prec f(x)$, then $x \in M$.

Given $P \xrightarrow{f} Q \xrightarrow{g} N$, the map f is *birhizomic at* g if f is of finite presentation in the category of monoids and, for every factorization

$$P \xrightarrow{h} M \xrightarrow{p} N$$

of gf with p infinitesimal, there is a unique $u : Q \to M$ such that $pu = g$ and $uf = h$.

A family $P \to Q_i$ is a *birhizomic cover* if, for every valuative monoid V and every map $v : P \to V$, there is some i and a factorization

$$P \longrightarrow Q_i \longrightarrow V$$

of v such that $P \to Q_i$ is birhizomic at $Q_i \to V$. For an object $P \to Q$ of A_P, covering families in the birhizomic topology are defined by the same condition with Q in place of P. In this paper we use only the valuative factorization part of the definition.

2.2 Faces and the stratum category

For $x \in P$, define

$$F_x := \{z \in P \mid \exists w \in P,\ n \geq 1 \text{ such that } z + w = nx\}.$$

This is the smallest face of P containing x; see Lemma 3.1.

The object $b_x : P \to Q_x = P[-x]^\sharp$ is the initial map under P killing x. In Proposition 3.2 we give an explicit model for Q_x and prove that the kernel of $P \to Q_x$ is exactly F_x.

Let A_P be the category of monoid homomorphisms $P \to Q$ under P, and let $B_P \subset A_P$ be the full subcategory on the objects b_x. We write $j : B_P \hookrightarrow A_P$ for the inclusion. On opposite categories we have

$$i : B_P^{\mathrm{op}} \hookrightarrow A_P^{\mathrm{op}}.$$

The site A_P^{op} carries the birhizomic topology J. The site B_P^{op} carries the trivial topology. We identify presheaves on B_P^{op} with covariant functors $B_P \to \mathbf{Ab}$, and similarly for A_P^{op}. Under this identification, the inverse-image functor is

$$\pi^{-1} = a \circ \mathrm{Lan}_j,$$

where $\mathrm{Lan}_j : \mathbf{Ab}^{B_P} \to \mathbf{Ab}^{A_P}$ is left Kan extension along j, and a is sheafification for the birhizomic topology on A_P^{op}. The direct-image functor is restriction:

$$\pi_* = j^* \circ \iota,$$

where $\iota : \mathrm{Sh}(A_P^{\mathrm{op}}) \hookrightarrow \mathrm{PSh}(A_P^{\mathrm{op}})$ is the inclusion.

3 The face-stratum category B_P

Lemma 3.1. *For every $x \in P$, the subset $F_x \subset P$ is the smallest face of P containing x.*

Proof. First, F_x is a submonoid. Indeed, $0 + x = x$, so $0 \in F_x$. If $z_1 + w_1 = n_1 x$ and $z_2 + w_2 = n_2 x$, then

$$(z_1 + z_2) + (w_1 + w_2) = (n_1 + n_2)x,$$

so $z_1 + z_2 \in F_x$. Also $x + 0 = 1x$, so $x \in F_x$.

If $a + b \in F_x$, say $a + b + w = nx$, then

$$a + (b + w) = nx, \qquad b + (a + w) = nx,$$

so $a, b \in F_x$. Thus F_x is a face.

Finally, if G is any face containing x, then $nx \in G$ for all $n \geq 1$. If $z + w = nx$, then since $nx \in G$ and G is a face, $z \in G$. Hence $F_x \subset G$. Therefore F_x is the smallest face containing x. □

Proposition 3.2. *Let $x \in P$. Let*

$$H := P^{\mathrm{gp}}/\langle x \rangle,$$

let $T \subset H$ be its torsion subgroup, and let

$$G := H/T.$$

Let $\rho : P \to G$ be the composite map. Define

$$S := \{s \in G \mid \exists m \geq 1 \text{ with } ms \in \rho(P)\},$$

and let

$$U := S \cap (-S).$$

Then:

(i) $Q_x := S/U$ is an integral, saturated, sharp monoid.

(ii) The composite map

$$\varphi_x : P \xrightarrow{\rho} S \longrightarrow S/U = Q_x$$

sends x to 0 and is initial among maps from P to integral, saturated, sharp monoids which send x to 0.

(iii) The kernel of φ_x is exactly F_x.

Thus Q_x is canonically isomorphic to $P[-x]^\sharp$.

Proof. We first prove (i). The set S is a submonoid of G: if $s, t \in S$, choose $m, n \geq 1$ with $ms, nt \in \rho(P)$. Then

$$mn(s+t) = n(ms) + m(nt) \in \rho(P),$$

so $s+t \in S$. By definition, S is saturated in G: if $ns \in S$, then for some $m \geq 1$, $mns \in \rho(P)$, hence $s \in S$.

The subset $U = S \cap (-S)$ is the subgroup of units of S. Form the quotient monoid $Q_x = S/U$. It is a submonoid of the group G/U, hence integral.

We check saturation. Suppose $n\overline{s} = \overline{t}$ in Q_x, with $t \in S$. Then in G/U,

$$ns - t \in U.$$

Since $U \subset S$, we have

$$ns = t + (ns - t) \in S.$$

Because S is saturated, $s \in S$. Hence $\overline{s} \in Q_x$.

We check sharpness. Suppose $\overline{s} + \overline{t} = 0$ in Q_x, with $s, t \in S$. Then $s + t \in U$, so $-(s+t) \in S$. Since $t \in S$, we have

$$-s = t + (-(s+t)) \in S.$$

Thus $s \in S \cap (-S) = U$, so $\overline{s} = 0$. Therefore Q_x is sharp.

For (ii), first note that $\rho(x) = 0$ in H, hence in G, so $\varphi_x(x) = 0$.

Let $f : P \to R$ be a map to an integral, saturated, sharp monoid R with $f(x) = 0$. Since R is integral, f extends uniquely to a group homomorphism

$$f^{\mathrm{gp}} : P^{\mathrm{gp}} \longrightarrow R^{\mathrm{gp}}.$$

Because $f(x) = 0$, this factors through $H = P^{\mathrm{gp}}/\langle x \rangle$. The group R^{gp} is torsion-free: if $ng = 0$, saturation gives $g, -g \in R$, and sharpness gives $g = 0$. Therefore f^{gp} factors uniquely through $G = H/T$. Let

$$\alpha : G \longrightarrow R^{\mathrm{gp}}$$

be the resulting homomorphism.

If $s \in S$, choose $m \geq 1$ with $ms = \rho(p)$ for some $p \in P$. Then

$$m\alpha(s) = f(p) \in R.$$

Since R is saturated, $\alpha(s) \in R$. Thus $\alpha(S) \subset R$.

If $u \in U$, then $\alpha(u) \in R$ and $-\alpha(u) \in R$. Hence $\alpha(u)$ is a unit in R. Since R is sharp, $\alpha(u) = 0$. Therefore α factors uniquely through $S/U = Q_x$, giving a monoid homomorphism

$$\overline{\alpha} : Q_x \longrightarrow R$$

such that $\overline{\alpha} \circ \varphi_x = f$.

Uniqueness follows because P^{gp} generates G, hence Q_x^{gp} is generated by the image of P. This proves the universal property.

For (iii), first suppose $z \in F_x$. Then $z + w = nx$ for some $w \in P$, $n \geq 1$. Applying ρ, we get

$$\rho(z) + \rho(w) = n\rho(x) = 0.$$

Thus $\rho(z) = -\rho(w) \in S \cap (-S) = U$, so $\varphi_x(z) = 0$. Hence $F_x \subset \ker(\varphi_x)$.

Conversely, suppose $\varphi_x(z) = 0$. Then $\rho(z) \in U$, so $-\rho(z) \in S$. Hence there exist $m \geq 1$ and $p \in P$ such that

$$m(-\rho(z)) = \rho(p)$$

in G. Equivalently, the class of $-mz - p$ in $H = P^{\mathrm{gp}}/\langle x \rangle$ is torsion. Therefore there is $\ell \geq 1$ such that

$$\ell(-mz - p) \in \langle x \rangle$$

in P^{gp}. Thus for some integer c,

$$-\ell mz - \ell p = cx.$$

Put $B = \ell m > 0$ and $p' = \ell p \in P$. Then

$$Bz + p' = -cx.$$

Let $D = -c$. If $D > 0$, then

$$Bz + p' = Dx,$$

so $Bz \in F_x$. Since F_x is a face and $Bz = z + \cdots + z$, it follows that $z \in F_x$.

If $D \leq 0$, then

$$Bz + p' + |D|x = 0$$

in P. Since P is sharp, a sum of elements of P can be zero only if each summand is zero. Hence $Bz = 0$, and sharpness gives $z = 0$. Thus again $z \in F_x$.

Therefore $\ker(\varphi_x) = F_x$. □

Corollary 3.3. *For $x, y \in P$, the set*

$$\mathrm{Hom}_{B_P}(b_x, b_y)$$

is a singleton if $F_x \subset F_y$, and empty otherwise. Hence B_P is equivalent to the poset of faces of the form F_x.

Proof. By the universal property of Q_x, a map $Q_x \to Q_y$ under P exists if and only if the composite

$$P \longrightarrow Q_y$$

kills x. By Proposition 3.2, this is equivalent to $x \in F_y$. Since F_y is a face, $x \in F_y$ is equivalent to $F_x \subset F_y$. When such a map exists, it is unique by the same universal property. □

4 Exactness of the inverse-image functor

We keep the identification $\mathrm{PSh}(B_P^{\mathrm{op}}) \simeq \mathbf{Ab}^{B_P}$ and $\mathrm{PSh}(A_P^{\mathrm{op}}) \simeq \mathbf{Ab}^{A_P}$. Under this identification,

$$\pi^{-1} = a \circ \mathrm{Lan}_j .$$

Proposition 4.1. *For every monoid P, the functor*

$$\pi^{-1} : \mathrm{PSh}(B_P^{\mathrm{op}}, \mathbf{Ab}) \longrightarrow \mathrm{Sh}(A_P^{\mathrm{op}}, \mathbf{Ab})$$

is exact.

Proof. Let $F : B_P \to \mathbf{Ab}$ be a presheaf. For an object $X = (f : P \to Q)$ of A_P, the left Kan extension is computed by

$$(\mathrm{Lan}_j F)(X) = \varinjlim_{(b_x, u : b_x \to X)} F(b_x),$$

where the colimit is over the comma category $(j \downarrow X)$.

By Proposition 3.2, a map $u : b_x \to X$ under P exists, uniquely, if and only if $f(x) = 0$. Hence $(j \downarrow X)$ is equivalent to the poset of those $x \in P$ such that $f(x) = 0$, with a morphism $x \to x'$ exactly when $F_x \subset F_{x'}$.

This poset is filtered. It is nonempty because $f(0) = 0$. If $f(x) = f(x') = 0$, then $f(x + x') = 0$. Also $x \in F_{x+x'}$ and $x' \in F_{x+x'}$, because

$$x + x' = 1 \cdot (x + x'),$$

so $F_x, F_{x'} \subset F_{x+x'}$. Thus $x + x'$ is a common upper bound. Since the category is a poset, parallel arrows are automatically equal. Therefore $(j \downarrow X)$ is filtered.

Filtered colimits of abelian groups are exact: representatives may be moved to a common index, and exactness is checked there. Therefore Lan_j is exact.

Sheafification for abelian presheaves on a small site is exact; see [1, Exp. II, Prop. 3.4]. Hence

$$\pi^{-1} = a \circ \mathrm{Lan}_j$$

is exact. □

5 The maximal-sieve criterion

Let J denote the birhizomic topology on A_P^{op}. A *stratum* is an object $b_x \in B_P$, viewed as an object of A_P^{op}.

Lemma 5.1. *Suppose that for every $x \in P$, the monoid $Q_x = P[-x]^\sharp$ is valuative. Then every J-covering sieve of every stratum b_x in A_P^{op} is maximal.*

Proof. It is enough to show that every birhizomic covering family of a stratum generates the maximal sieve, because every covering sieve contains the sieve generated by a covering family.

Fix $x \in P$. Let

$$s_i : Q_x \longrightarrow Q_i$$

be a birhizomic covering family of the stratum b_x. In A_P^{op}, this gives arrows

$$U_i = (P \to Q_i) \longrightarrow b_x.$$

By assumption, Q_x is valuative. Apply the defining valuative factorization property of a birhizomic cover with $V = Q_x$ and $v = \mathrm{id}_{Q_x}$. There exists some i and a map

$$r : Q_i \longrightarrow Q_x$$

such that

$$r \circ s_i = \mathrm{id}_{Q_x}.$$

Thus s_i is a split monomorphism under Q_x.

Now let

$$U = (P \to Z) \longrightarrow b_x$$

be any arrow into b_x in A_P^{op}. Equivalently, it is a map

$$t : Q_x \longrightarrow Z$$

under P. Define

$$h := t \circ r : Q_i \longrightarrow Z.$$

Then h is a map under P, and the corresponding arrow

$$U \longrightarrow U_i$$

in A_P^{op} makes the triangle commute:

$$U \longrightarrow U_i \longrightarrow b_x$$

is the original arrow $U \to b_x$. Hence every arrow into b_x factors through the covering arrow $U_i \to b_x$. The sieve generated by the covering family is therefore maximal, and any covering sieve containing it is maximal. □

We now state the main sufficient condition.

Every J-covering sieve of every stratum $b_x \in B_P$ is maximal. (Max)

Theorem 5.2. *Assume* (Max). *Then for every presheaf of abelian groups F on B_P^{op}, the unit*

$$F \longrightarrow R\pi_*\pi^{-1}F$$

is a quasi-isomorphism. Moreover, π_* is exact.

Proof. First consider the underived unit

$$F \longrightarrow \pi_* \pi^{-1} F.$$

Since $j : B_P \hookrightarrow A_P$ is fully faithful, the canonical map

$$F \longrightarrow j^* \operatorname{Lan}_j F$$

is an isomorphism. Explicitly, for $b \in B_P$, the comma category $(j \downarrow b)$ has terminal object (b, id_b), so

$$(\operatorname{Lan}_j F)(b) \cong F(b).$$

Let H be any presheaf on A_P^{op}. We claim that under (Max),

$$(aH)(b) = H(b)$$

for every stratum $b \in B_P$. Indeed, in the plus construction,

$$H^+(b) = \varinjlim_{R \in J(b)} \operatorname{Match}(R, H).$$

By (Max), the only covering sieve of b is the maximal sieve. A matching family over the maximal sieve is uniquely determined by its value at id_b, so

$$\operatorname{Match}(\text{maximal sieve}, H) \cong H(b).$$

Thus $H(b) \to H^+(b)$ is an isomorphism. Applying the same argument to H^+, we obtain $(aH)(b) = H(b)$.

Taking $H = \operatorname{Lan}_j F$, we get

$$(\pi_* \pi^{-1} F)(b) = (a \operatorname{Lan}_j F)(b) = (\operatorname{Lan}_j F)(b) \cong F(b).$$

The underived unit is therefore an isomorphism of presheaves on B_P^{op}.

Next we prove that π_* is exact. Since π_* is a right adjoint, it is left exact. It remains to show that it preserves epimorphisms.

Let $\varphi : G \to H$ be an epimorphism of abelian sheaves on A_P^{op}. Fix a stratum $b \in B_P$ and a section $y \in H(b)$. Consider the sieve

$$S_y := \{\, \alpha : U \to b \mid \alpha^* y \text{ lifts to } G(U) \,\}.$$

Because φ is an epimorphism of sheaves, it is locally surjective, so S_y is a covering sieve. By (Max), S_y is maximal. Hence $\mathrm{id}_b \in S_y$, so y lifts to $G(b)$. Thus

$$G(b) \longrightarrow H(b)$$

is surjective for every stratum b. Therefore $\pi_* \varphi$ is an epimorphism of presheaves on B_P^{op}.

Thus π_* is left exact and preserves epimorphisms. Since it is additive, it is exact. Consequently

$$R^q \pi_* = 0 \qquad (q \geq 1).$$

Finally, for any presheaf F on B_P^{op}, the complex $R\pi_* \pi^{-1} F$ has cohomology only in degree 0, and its degree-0 cohomology is $\pi_* \pi^{-1} F$. The degree-0 map is the underived unit, which we proved is an isomorphism. Therefore

$$F \longrightarrow R\pi_* \pi^{-1} F$$

is a quasi-isomorphism. □

6 Rank-one monoids

Lemma 6.1. *If* $\operatorname{rank}(P^{\mathrm{gp}}) = 1$*, then* P *is valuative.*

Proof. Since P^{gp} has rank one, it is nonzero and torsion-free. Choose $0 \neq p \in P$. Let $g \in P^{\mathrm{gp}}$. In the one-dimensional $\mathbf{Q}$-vector space $P^{\mathrm{gp}} \otimes \mathbf{Q}$, the elements g and p are linearly dependent. Hence there exist integers $m \geq 1$ and k such that

$$mg = kp.$$

If $k \geq 0$, then $mg \in P$, so saturation gives $g \in P$. If $k < 0$, then

$$m(-g) = (-k)p \in P,$$

so saturation gives $-g \in P$. If $k = 0$, then $mg = 0$, hence $g = 0$ because P^{gp} is torsion-free. Thus for every $g \in P^{\mathrm{gp}}$, either $g \in P$ or $-g \in P$, so P is valuative. □

Lemma 6.2. *Assume* $\operatorname{rank}(P^{\mathrm{gp}}) = 1$*. Then:*

(i) $Q_0 = P$.

(ii) If $x \neq 0$*, then* $Q_x = 0$.

(iii) Every stratum Q_x *is valuative.*

(iv) B_P *has exactly two objects up to canonical isomorphism, namely* $b_0 : P \to P$ *and* $b_1 : P \to 0$*, with a single non-identity arrow* $b_0 \to b_1$.

Proof. For $x = 0$, the explicit model of Proposition 3.2 gives $Q_0 = P$.

Let $x \neq 0$. Since P^{gp} has rank one, the quotient

$$P^{\mathrm{gp}}/\langle x \rangle$$

is a torsion group. Hence its torsion-free quotient G is zero. The model of Proposition 3.2 therefore gives $Q_x = 0$.

The monoid P is valuative by Lemma 6.1, and the zero monoid is valuative because its associated order is trivially total. Hence every stratum Q_x is valuative.

Finally, if $x \neq 0$, then $Q_x = 0$, so all nonzero x give the same object $b_1 : P \to 0$. By Corollary 3.3, there is a unique arrow $b_0 \to b_1$ because $F_0 = \{0\} \subset P = F_x$ for $x \neq 0$, and there is no arrow $b_1 \to b_0$ because $F_x = P \not\subset F_0$. Thus B_P is the two-object poset $b_0 \to b_1$. □

Theorem 6.3. *Let* P *be a monoid with* $\operatorname{rank}(P^{\mathrm{gp}}) = 1$*. Then for every presheaf of abelian groups* F *on* B_P^{op}*, the unit*

$$F \longrightarrow R\pi_*\pi^{-1}F$$

is a quasi-isomorphism.

Proof. By Lemma 6.2, every stratum Q_x is valuative. Lemma 5.1 therefore implies condition (Max). The result then follows from Theorem 5.2. □

7 Finitely generated valuative monoids

Lemma 7.1. *Let P be a finitely generated valuative monoid. Then for every $x \in P$, the monoid*

$$Q_x = P[-x]^\sharp$$

is valuative.

Proof. Use the explicit model of Proposition 3.2. We have

$$Q_x = S/U \subset G/U,$$

where G is a quotient of P^{gp}, S is the saturation of the image of P, and $U = S \cap (-S)$. Since S generates G, the group completion of Q_x is

$$Q_x^{\mathrm{gp}} \cong G/U.$$

Let $\gamma \in Q_x^{\mathrm{gp}}$. Choose a lift $g \in P^{\mathrm{gp}}$ of γ. Since P is valuative, either $g \in P$ or $-g \in P$. If $g \in P$, then its image in G lies in $\rho(P) \subset S$, hence $\gamma \in Q_x$. If $-g \in P$, then similarly $-\gamma \in Q_x$.

Thus for every $\gamma \in Q_x^{\mathrm{gp}}$, either $\gamma \in Q_x$ or $-\gamma \in Q_x$. Since Q_x is sharp, this order is antisymmetric. Hence Q_x is valuative. □

Theorem 7.2. *Let P be a finitely generated valuative monoid. Then for every presheaf of abelian groups F on B_P^{op}, the unit*

$$F \longrightarrow R\pi_*\pi^{-1}F$$

is a quasi-isomorphism.

Proof. By Lemma 7.1, every stratum Q_x is valuative. Lemma 5.1 gives condition (Max). Theorem 5.2 then applies. □

8 Open points

The following points remain open.

(1) **General non-valuative monoids.** The proof of (Max) uses valuativity of the strata. For non-valuative monoids, such as $P = \mathbf{N}^2$, the argument fails. No proof or counterexample is known for the general case.

(2) **Necessity of the maximal-sieve condition.** We proved only the sufficiency direction:

$$\text{(Max)} \quad \Longrightarrow \quad F \to R\pi_*\pi^{-1}F \text{ is a quasi-isomorphism for all } F.$$

The converse is not proved here.

(3) **Higher-rank non-finitely generated valuative monoids.** The rank-one theorem covers all rank-one monoids, including non-finitely generated ones. The valuative theorem above is stated for finitely generated valuative monoids. Extension to arbitrary higher-rank valuative monoids is not addressed.

(4) **Higher cohomology and descent spectral sequences.** In the positive regimes proved here, higher direct images vanish because π_* is exact. The general behavior of higher cohomology and descent spectral sequences remains open.

References

[1] M. Artin, A. Grothendieck, J.-L. Verdier, *Théorie des Topos et Cohomologie Étale des Schémas (SGA 4)*, Tome 1, Lecture Notes in Mathematics 269, Springer, 1972. Exposé II, Proposition 3.4.